Soil Microbiome Needs Assessment

Interviewer Notes | One copy per interviewer per interview | Confidential

How this document is used

How this document is used: *Each interviewer completes their own copy of this document during the interview. Where two or more interviewers are present, each captures notes independently. After the interview, interviewers compare and reconcile their notes, then one person transfers the agreed, amalgamated summary into the correct interviewee column of the Soil Needs Assessment Summary Matrix. Classification of data elements (Must / Should / Optional / Inferred) is done at the matrix stage, not here.*

Step 1 — During the interview: Each interviewer types raw notes into their own copy of this document. Capture what the interviewee actually says — exact tool names, units, and verbatim pain points are especially valuable.

Step 2 — Immediately after: Interviewers compare notes and flag any discrepancies or gaps. Agree on a single consolidated account of what was said for each question.

Step 3 — Transfer to matrix: One interviewer enters the consolidated notes into the interviewee column of the Summary Matrix. Use the 'Key data elements' boxes at the end of each section (below) as a staging area before transferring.

Step 4 — After all interviews: Once all interviewee columns are populated, use the matrix to assign classifications and complete the Theme Synthesis sheet.

Interview Details

Date	
Interview ID	*e.g. INT-01*
Research domain / project type	
Institution	*e.g. university, government, NGO*
Interviewer(s)	
Author	

Pre-Interview Checklist

- ☐ Reviewed interviewee's publications or project page
- ☐ Identified most relevant sections for their role
- ☐ Confirmed purpose of interview with interviewee
- ☐ Confirm consent for note-taking given

Questions

1. Scope and audience

> **Listen for:** *Key objectives · intended users and data consumers · land management relevance · comparability needs · regulatory or funding drivers. Note nuances that may affect how data must be captured, shared, or compared.*

What are the primary questions or objectives of your project (e.g. surveillance, research, monitoring a specific pathogen or target)?

Notes

Who are your partners and/or collaborators?

Notes

What decisions (if any) will be informed by your data? (e.g. agricultural management, risk assessment, policy, remediation)

Notes

What land management practices are relevant to your project (e.g. tillage, grazing, irrigation, mowing, crop rotation, restoration)?

Notes

What level of comparability do you need? Is it within a project, across sites, across time, or across organisations?

Notes

Who ultimately consumes or interprets the data (e.g., modelers, regulators, farmers)?

Notes

Key data elements / terms to transfer to the Summary Matrix

2. Sampling strategy

Listen for: *Sample types · soil fractions targeted · spatial/temporal coverage · control strategy. Identify nuances in controls, replicates, and derived sample types that will need to be represented in a data model.*

Is there a sample plan? Is the sample plan, experimental design, or any other material accessible?

Notes

What kinds of samples are collected?

Notes

Are only soil samples collected or are there additional sample types (e.g. host, environmental, food, wastewater)?

Notes

What types of soil samples are specifically collected (e.g. soil cores, rhizosphere, bulk soil, aggregates, water-adjacent soil)?

Notes

What is the sampling frequency and/or duration (e.g. one-time, seasonal, longitudinal, event-triggered)?

Notes

What are your collection methods and devices (e.g. soil corers, augers, shovels, syringes, lysimeters, pore-water samplers)? Are any specific to a sample type, depth, or horizon?

Notes

What unique identifiers are assigned to samples? (e.g. barcodes, lab IDs, field IDs, site-date-replicate codes)

Notes

How are identifiers tracked across the full sample lifecycle?

Notes

How are identifiers linked between parent samples and derived subsamples or aliquots?

Notes

Are you incorporating any previously collected samples or existing datasets?

Notes

Are you collecting any controls? What types (e.g. biological, technical)?

Notes

Key data elements / terms to transfer to the Summary Matrix

3. Geographical & Site Context

Listen for: *Push for precision — 'a field in Wales' is not useful, a GPS coordinate or shapefile reference is. Probe pre-sampling history (planned and unplanned) as this directly affects microbial community composition.*

What environmental site does your sampling area occupy?

Notes

What are the sites or land uses in the immediate vicinity that could influence your samples (proximal environments)?

Notes

Is your site exposed to any planned pre-sampling treatments or interventions (e.g. fertiliser, manure, pesticides/herbicides, mechanical events, experimental interventions)?

Notes

Has your site been exposed to any unplanned major environmental or anthropogenic events (e.g. fire, flood, drought, storm damage, animal intrusion)?

Notes

At what level of spatial specificity are sampling sites described (e.g. GPS point, grid reference, polygon boundary, geospatial layer)? Is this standardised or free text?

Notes

What is the spatial extent of the sampling area and how is it represented (e.g. GPS points, shapefiles, transects, watershed boundary)?

Notes

Key data elements / terms to transfer to the Summary Matrix

4. Physico-Chemical & Biological Context

Listen for: *Avoid general answers — push for specific properties, measurement units, and analytical methods. For each measurement, probe: what is the value, what are the units, what is the method? Free-text descriptions are not interoperable.*

How is the soil classified or characterised (e.g. texture, structure, horizon designation, rockiness, colour)?

Notes

At what depth(s) are soil samples or measurements taken, and how are soil horizons or layers defined?

Notes

What physico-chemical data is collected to describe soil properties (e.g. pH, soil organic matter/carbon, moisture content, salinity, cation exchange capacity, bulk density, redox potential)?

Notes

What nutrients or compounds are being measured (e.g. nitrogen species, phosphorus fractions, potassium, micronutrients, organic pollutants, microplastics)?

Notes

Are there other biological properties measured beyond sequencing (e.g. biomass, enzyme activity, respiration, gas flux)?

Notes

Are meteorological or microclimate data recorded (e.g. air/soil temperature, rainfall, humidity, solar radiation)?

Notes

Key data elements / terms to transfer to the Summary Matrix

5. Specimen receipt and storage

Listen for: *If samples move between institutions, ask specifically how information is transferred and whether chain-of-custody is formally recorded. Cold chain and storage conditions directly affect microbial community composition.*

Are there formal receipt procedures when samples arrive at the lab (e.g. documentation, quality checks, chain-of-custody)?

Notes

Are environmental conditions during transport recorded (e.g. temperature throughout transit, duration of transit)?

Notes

How are samples stored and for how long (e.g. temperature, duration, location tracking)?

Notes

How is contamination tracked and documented, and at which stage is it most likely to occur (e.g. pre-sampling, transport, sequencing)?

Notes

Is there a system to track sample integrity over time (e.g. freeze-thaw cycles, degradation)?

Notes

Key data elements / terms to transfer to the Summary Matrix

6. Specimen processing

Listen for: *Extraction is a critical step — push for specific kit names and versions, input masses, and inhibitor-removal steps. Ask about SOPs and whether these are shared across lab sites. These details are often absent from published methods but essential for data reuse.*

Are samples further subsampled, pooled, or otherwise derived once received?

Notes

How is sample mass or volume quantified at each processing step? Is any normalisation applied?

Notes

What is the typical input mass of soil used for extraction? Does this vary by sample type or horizon?

Notes

Are samples subjected to any special processing prior to nucleic acid extraction (e.g. fractionation, filtering, enrichment, culturing)?

Notes

How do processing steps potentially bias microbial composition?

Notes

Is any culturing or isolation performed? If so, what media and growth conditions are used?

Notes

What DNA or RNA extraction methods are used (e.g. magnetic bead-based, enzymatic lysis, mechanical disruption)? Kit name and version?

Notes

How is the quality and quantity of extracted nucleic acid assessed (e.g. Qubit fluorometry, NanoDrop, Bioanalyzer/TapeStation, gel electrophoresis)?

Notes

Are extraction protocols standardised across sites or labs?

Notes

Key data elements / terms to transfer to the Summary Matrix

7. Library preparation

Listen for: *Assay choice (amplicon vs. shotgun) fundamentally determines what contextual data is most important to capture. For amplicon studies, the specific target region is a critical variable for comparability. Note whether protocols are in-house or published.*

What are the primary sequencing assay(s) being used (e.g. WGS, WRS, amplicon, 16S, 18S)?

Notes

Is library preparation performed in-house or by an external partner or core facility?

Notes

What library preparation or enrichment methods are applied (e.g. tagmentation, ligation-based, PCR amplification, size selection)?

Notes

Are commercial library kits or in-house protocols used?

Notes

What primers are used?

Notes

Do you have a shareable or open access protocol?

Notes

What sequencing controls are included at the library preparation stage (e.g. mock community standards, synthetic spike-ins, endogenous controls, technical replicates)?

Notes

Key data elements / terms to transfer to the Summary Matrix

8. Sequencing and Bioinformatics

Listen for: *Probe for specific tools, database versions, and parameter choices — often under-reported but critical for reproducibility. Note where taxonomy is assigned and which database is used: a key point of divergence across projects.*

Is sequencing performed in-house or with an external provider?

Notes

What sequencing instruments and platforms are used?

Notes

Can you walk through your bioinformatics workflow, from raw sequence data to final outputs?

Notes

What tools, reference databases, and custom scripts are used within the pipeline (e.g. Trimmomatic, DADA2, Kraken2, MetaPhlAn)?

Notes

How is taxonomy assigned, and which database is used?

Notes

Are metagenome-assembled genomes (MAGs) or binning approaches used?

Notes

How is quality control applied and recorded throughout the pipeline?

Notes

Is your workflow open access or shareable?

Notes

Key data elements / terms to transfer to the Summary Matrix

9. Data Management

> **Listen for:** *Listen for workarounds, manual reformatting steps, and information lost in transit — these flag where a standard would add the most value. Ask about systems and tools, and how these integrate with collaborators' workflows.*

How is contextual data captured (e.g. spreadsheets, LIMS, REDCap)?

Notes

Are any metadata or data standards applied (e.g. MIxS/MIMARKS/MIMS, Darwin Core, ISA-Tab, INSPIRE, DataCite)?

Notes

Where is the data being stored (e.g. locally, public repositories, network databases)?

Notes

Is there a plan to submit data to a public repository?

Notes

What are the data sharing workflows — where does it need to go, how does it get there?

Notes

During data flow, will the data need to be transformed?

Notes

Are there governance, policy, or privacy considerations affecting data sharing?

Notes

Is there a formal Data Management Plan (DMP) in place?

Notes

How is data versioning or provenance tracked?

Notes

Key data elements / terms to transfer to the Summary Matrix

10. Associated Data Types

Listen for: *Examples: plant community surveys, biochemical assays, environmental sensor data, clinical/agricultural records. Aim is not to capture everything — but to know what exists so it can be referenced or linked in the data standard.*

Are there additional data types associated with your samples that we have not yet discussed?

Notes

Which of these would be most useful to reference or flag alongside the primary sequencing data in a data standard?

Notes

Are there plans to formally link or integrate these datasets with the genomic data?

Notes

Key data elements / terms to transfer to the Summary Matrix

11. Future Work

Listen for: *Encourage reflection beyond their current project. Ask what they would change if starting again, and what they wish existed. This section surfaces emerging requirements for interoperability and gaps in current community resources.*

What data types would you want to prioritise collecting or integrating in the future?

Notes

What are your main pain points with contextual data collection or management today?

Notes

Are there emerging technologies or methods you anticipate adopting that might change your data needs?

Notes

Are there existing data standards or community resources that are close to meeting your needs but fall short in specific ways?

Notes

Key data elements / terms to transfer to the Summary Matrix

Post-Interview Notes

Complete while the interview is fresh, before comparing notes with co-interviewers. These notes feed directly into Step 2 (reconciliation) and Step 3 (transfer to matrix).

Overall impressions / standout themes

Notes

Pain points mentioned (verbatim where possible)

Notes

Data elements the interviewee was emphatic about

Notes

Areas of ambiguity or contradiction — flag for discussion with co-interviewer

Notes

Terminology or controlled vocabulary terms used by the interviewee

Notes

Tools / databases / standards mentioned

Notes

Transfer Checklist

Complete once notes have been reconciled with co-interviewers and transferred to the Summary Matrix.

- ☐ Notes compared and reconciled with co-interviewer(s)
- ☐ Consolidated notes entered into the correct interviewee column of the Summary Matrix
- ☐ Key data elements extracted from each section's amber box
- ☐ Pain points and verbatim quotes noted in the matrix
- ☐ Any flagged items or follow-ups recorded
- ☐ Interviewer initials and transfer date recorded in the matrix